

COLONIA DEL SACRAMENTO, Uruguay

WMO: 865600

Lat: 34.452S

Lon: 57.767W

Elev: 66

StdP: 14.66

Time Zone: -3.00 (W03)

Period: 94-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.1	41.5	28.3	22.3	44.3	31.1	25.4	46.0	33.9	49.2	31.1	48.7	7.3	60	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.6	88.2	73.4	85.9	72.5	83.9	71.7	76.6	84.6	75.2	82.7	73.8	80.6	9.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.1	127.7	81.7	72.8	121.8	80.2	71.5	116.5	78.5	40.2	84.7	38.8	83.0	37.5	80.6	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.7	25.9	22.6	34.3	95.5	1.3	3.0	33.4	97.7	32.6	99.4	31.9	101.1	31.0	103.3		
(5)			32.5	80.7	1.8	1.7	31.2	81.9	30.1	82.9	29.1	83.8	27.7	85.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	74.0	73.0	71.0	64.4	59.5	53.7	52.2	55.9	57.8	63.2	67.8	72.4		
	DBStd	9.24	4.42	4.36	4.59	4.78	5.44	5.99	5.97	6.76	5.41	5.14	5.36	5.16		
	HDD50	85	0	0	0	0	1	26	40	14	3	1	0	0		
	HDD65	1666	1	2	7	66	187	344	403	299	226	96	29	6		
	CDD50	5081	744	643	650	432	295	137	108	197	237	410	534	694		
	CDD65	1186	280	225	191	48	16	5	5	17	10	40	113	236		
	CDH74	7238	2095	1393	910	132	25	6	17	106	63	170	642	1679		
	CDH80	1685	539	315	165	13	2	0	1	27	10	30	142	441		
(14) Wind	WSAvg	10.7	10.2	10.8	10.6	10.9	9.3	10.7	10.5	11.0	10.9	11.0	11.1	11.3		
(15) Precipitation	PrecAvg	45.2	4.4	5.0	4.7	4.0	3.0	2.7	2.8	3.1	4.4	4.5	4.2	4.2		
	PrecMax	68.7	9.4	13.5	19.1	9.7	12.0	9.1	6.2	8.1	7.7	12.2	12.9	14.0		
	PrecMin	25.3	1.1	1.2	1.0	0.5	0.4	0.4	0.4	0.6	0.4	1.1	0.1	0.1		
	PrecStd	10.7	2.2	3.4	3.6	2.6	2.3	1.8	1.5	2.1	2.0	2.8	2.6	3.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.9	89.9	88.2	81.7	77.1	74.2	76.5	83.6	80.7	84.4	88.2	91.4		
		MCWB	75.9	75.2	73.6	71.4	67.4	65.3	65.8	70.5	69.3	70.2	71.5	74.6		
	2%	DB	87.5	86.1	84.3	77.4	73.1	69.4	68.9	75.7	74.7	78.2	84.0	87.3		
		MCWB	73.9	73.1	72.9	67.9	65.3	64.1	62.0	65.7	64.3	66.9	68.6	72.3		
	5%	DB	85.2	83.6	81.2	74.9	70.0	65.6	63.8	70.8	70.0	75.0	80.5	84.4		
		MCWB	72.7	72.3	71.2	66.5	64.0	60.7	58.4	62.7	60.5	64.7	68.5	71.9		
	10%	DB	83.0	81.4	78.7	71.9	67.8	62.2	60.2	65.9	66.3	71.6	77.5	81.9		
		MCWB	71.6	71.2	70.2	65.8	62.6	58.6	55.6	58.4	58.5	63.5	66.6	70.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.0	77.4	77.8	73.6	69.7	67.0	65.8	70.6	71.3	72.9	74.8	78.0		
		MCDB	87.1	85.6	85.7	78.8	75.4	71.0	75.3	82.0	77.7	80.1	83.8	87.3		
	2%	WB	76.6	75.6	75.3	70.3	67.0	64.2	62.7	66.5	65.9	69.4	72.1	75.5		
		MCDB	84.3	83.5	80.7	75.0	71.1	67.9	67.9	74.4	73.8	74.9	79.5	84.3		
	5%	WB	74.9	74.0	73.4	68.5	65.0	62.0	59.4	63.4	62.1	67.3	70.0	73.6		
		MCDB	82.6	81.3	78.1	72.4	68.7	65.3	62.0	69.7	66.8	71.5	77.3	81.1		
	10%	WB	73.4	72.6	71.8	66.9	63.3	59.2	56.7	60.3	60.1	65.8	68.1	72.0		
		MCDB	80.5	79.5	76.8	70.7	66.9	61.4	59.4	64.6	64.2	70.0	75.8	79.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	14.3	13.3	12.0	11.3	10.5	11.4	13.1	13.4	13.6	15.9	15.7		
		MCDBR	18.7	17.7	16.9	16.1	14.1	14.6	16.0	19.6	19.7	19.2	20.4	20.2		
		MCWBR	11.0	10.5	9.5	10.5	9.8	10.2	9.9	11.6	12.1	11.2	11.0	12.4		
	5% WB	MCDBR	15.9	15.0	13.6	12.4	12.5	12.3	13.2	18.4	15.5	14.5	16.9	16.5		
		MCWBR	11.0	10.3	8.9	9.3	9.3	9.7	9.4	11.7	10.8	10.6	11.1	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.406	0.396	0.380	0.381	0.368	0.365	0.359	0.439	0.461	0.423	0.394	0.398		
	taud		2.400	2.446	2.475	2.412	2.422	2.423	2.412	2.134	2.090	2.268	2.386	2.398		
	Ebn at Noon		296	292	285	264	249	240	249	239	254	280	297	299		
	Edh at Noon		40	37	33	32	28	26	28	41	48	43	40	40		
(44) All-Sky Solar Radiation	RadAvg		2317	2012	1683	1220	850	719	749	1011	1392	1783	2161	2394		
	RadStd		137	188	157	137	70	71	82	101	95	165	158	124		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		+0.63	N/A	N/A	+1.15	+0.79	N/A	N/A	N/A	+191	+103	

Nomenclature: See separate page